



THE MAGIC GOOEY SCIENCE!

WHAT IS OOBLECK?

IS THIS MAGIC OR SCIENCE?

Have you ever seen something that looks like a liquid but turns SOLID when you touch it? What if you could actually WALK on goo without sinking? Sounds impossible, right?

Let's find out together! Get ready to discover one of the coolest science secrets hiding in your kitchen. This is going to be SO much fun!



IS THIS A SOLID OR LIQUID?



SOLIDS

Things that keep their shape:

- Jaggery
- Chapati
- Ice cube

LIQUIDS

Things that flow and pour:

- Milk
- Coconut oil
- Lemon Water

IN-BETWEEN

Sometimes solid, sometimes liquid:

- Wet beach sand
- Curd when stirred

THINK ABOUT IT!

What if something is BOTH a solid AND a liquid at the same time?

MEET THE NON-NEWTONIAN FLUID

the Tricky Liquid! A non-Newtonian fluid is a special liquid that changes how it behaves when you touch it. Touch it slowly – it flows like water. Touch it fast – it becomes hard like a solid! It's like magic, but it's really science!

Story Time: Little Arjun watches his dadi make chapati. He notices something cool – when dadi kneads the atta dough slowly, it feels smooth and soft. But when she presses it fast, it pushes back! That's just like our tricky Oobleck!





ONLY 2 THINGS FROM YOUR KITCHEN!



CORN FLOUR

Also called "maize (corn) flour." This white powder from your kitchen is the main ingredient. Ask mummy or dadi for help finding it!

WATER

Just plain water from your tap or bottle. We will mix it slowly with the corn flour to make magic happen!

FOOD COLOUR

Optional! Add any colour you like to make your Oobleck pretty. Fun fact: Oobleck is named after a Dr. Seuss story!

HOW TO MAKE OOBLECK



1

Mix 1 cup corn flour "maize (corn) flour." with half cup water. The magic ratio is 2:1 - two parts flour to one part water!

2

Add just 1 drop of food colour to make it fun and colourful. Pick your favourite colour - pink, green, or blue!

3

Mix everything slowly in a bowl with your hands. Keep stirring gently until it feels smooth and gooey. Your Oobleck is ready!

LET'S MAKE IT TOGETHER!

STEP 1: POUR CORN FLOUR

Start with 1 cup of corn flour "maize (corn) flour." in a big bowl. Make sure your bowl is wide enough to mix easily!

STEP 2: ADD WATER SLOWLY

Pour half cup of water slowly into the flour. Watch how it starts to change! Keep adding bit by bit.

STEP 3: ADD COLOUR & STIR

Add 1 drop of food colour if you like! Stir slowly with your hands. Feel the magic – your oobleck is ready to explore!



TEST 1 & 2: SLOW POUR AND FAST POKE

TEST 1: SLOW POUR

Tilt your bowl and let oobleck drip slowly. Watch it flow smoothly like honey or water. It acts just like a liquid when you're gentle with it!

WHAT HAPPENS?

When you move slowly, the tiny particles have time to slide past each other. That's why it flows! Slow equals liquid behavior.

TEST 2: FAST POKE

Now poke the oobleck really fast with your finger. Surprise! It feels hard like a solid. The faster you push, the harder it gets!



TEST 3 & 4: ROLL INTO A BALL AND STOP ROLLING



TEST 3: ROLL INTO A BALL

Roll the Oobleck quickly between your hands. It forms a ball and holds its shape! It acts like a **SOLID** when you move fast.

WHAT HAPPENS?

The faster you roll, the harder it feels! This is because the tiny particles jam together when pressure is applied quickly.

TEST 4: STOP ROLLING

Now stop rolling and hold still. Watch the ball slowly **MELT** back into gooey liquid in your hands! It flows like water again.

THE SCIENCE BEHIND OOBLECK

Why Does It Do This?

Tiny corn flour particles are packed together in water. When you press FAST, the particles jam together and act like a solid! When you move SLOW, the particles slide past each other and act like a liquid.

Think of it like a crowd in a Mumbai local train! When everyone pushes at once, nobody can move. But when people move slowly, they can slide past each other easily. That's exactly what happens inside oobleck!



SPOT THE NON- NEWTONIAN!



WET BEACH SAND

Walk on wet sand at Marina Beach, Goa, or Puri - it feels solid! But stand still and your feet sink in like liquid.

IDLI BATTER

Stir idli batter slowly and it flows. Stir it fast and it becomes thick and resists your spoon!

KETCHUP & FEVICOL

Shake the ketchup bottle to make it flow! Pull Fevicol slowly - it stretches. Pull fast - it snaps!

ATTA DOUGH

Knead atta slowly - it's smooth. Press hard and fast - it pushes back! Can you find these at home?

3 THINGS I LEARNED TODAY



1

Oobleck equals corn flour plus water! Just two simple ingredients from your kitchen can make this magical gooey substance. Mix them together and you have created something amazing!

2

It is a Non-Newtonian Fluid! This is a special type of liquid that does not follow the normal rules. It changes how it behaves depending on how you touch it. Pretty cool, right?

3

Press fast equals solid, move slow equals liquid! When you poke it quickly, it feels hard like a solid. When you move slowly, it flows like a liquid. Science is amazing!

Wonder and experiment more!

- Add more colour to your Oobleck?
- Put Oobleck on a speaker with music?
- Mix it with glitter?

Try it and tell us what happens!

WHAT IF YOU TRIED THIS?



Great Job Young Scientist!

You made something magical from your kitchen!
See you next time for more experiments!

Thank You!

